

AI-Based Resume Screening & ATS Optimization Platform

Abstract

The increasing number of job applications has made manual resume screening inefficient and time-consuming. This project presents an AI-based resume screening and ATS (Applicant Tracking System) optimization platform that automates candidate evaluation and ranking. The system leverages Natural Language Processing (NLP) and machine learning techniques to analyze resumes, extract key information such as skills, experience, and education, and match them with job requirements. It generates a relevance score and ranks candidates accordingly, improving hiring efficiency and accuracy. Such systems significantly reduce manual effort and enhance decision-making in recruitment processes.

Introduction

Recruitment processes today involve handling a large volume of resumes, making it difficult for recruiters to identify suitable candidates efficiently. Traditional screening methods are slow, prone to bias, and often result in missed opportunities. AI-powered resume screening systems address these challenges by automating the evaluation process.

This project focuses on building an intelligent platform that not only screens resumes but also optimizes them for ATS compatibility. The system analyzes resumes, matches them with job descriptions, and provides ranking and recommendations. Such platforms help recruiters save time and ensure fair and consistent candidate evaluation.

Tools Used

The following tools and technologies were used:

- Programming Language: Python
- Libraries:
 - Pandas, NumPy (data processing)
 - Scikit-learn (machine learning models)
 - NLTK / NLP libraries (text processing)
- Techniques:

- Resume Parsing
- Keyword Extraction
- Machine Learning Classification
- Similarity Matching (ATS scoring)
- Platform: Streamlit / Jupyter Notebook

These tools enable efficient data processing, feature extraction, and intelligent decision-making.

Steps Involved

1. Data Collection
 - Gathered resumes and job descriptions for training and testing
2. Data Preprocessing
 - Cleaned text data by removing noise, stopwords, and formatting issues
3. Resume Parsing
 - Extracted key details such as skills, education, and experience
4. Feature Extraction
 - Converted text into numerical vectors using NLP techniques
5. Model Development
 - Built machine learning models to match resumes with job descriptions
6. Ranking & Scoring
 - Assigned scores based on relevance and ATS compatibility
7. Deployment
 - Developed an interface to upload resumes and view ranked results

Conclusion

The AI-based resume screening and ATS optimization platform demonstrates how machine learning can transform recruitment processes. By automating resume analysis and ranking, the system reduces manual effort, improves efficiency, and enhances hiring decisions. Overall, the project highlights the growing importance of AI in modern recruitment systems.